



3D DIORAMA SLOT  
ART/COVER-FRONT.SVG

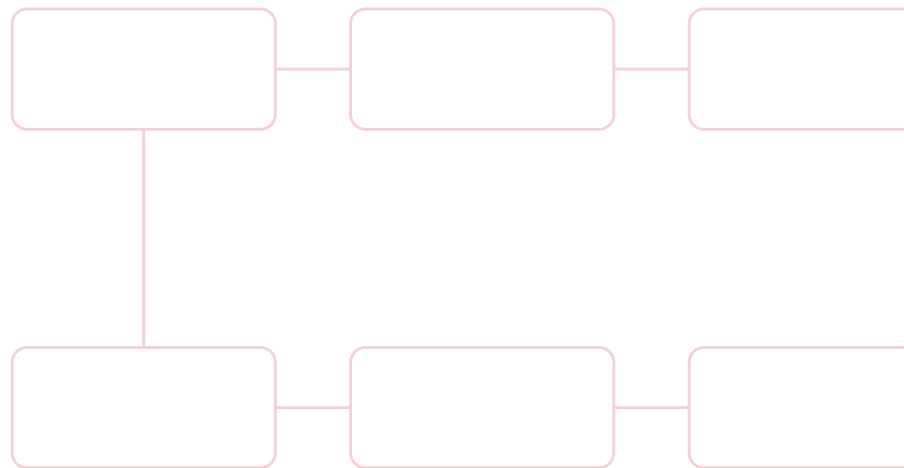
# AGENTIC AUTOML

*A printed system card • Vol. 01*

MIAMI UNIVERSITY • CSE 449 • SENIOR DESIGN PROJECT • 2026

AGENTIC-AUTOML.VERCEL.APP

*Welcome.*



Agentic AutoML automates the 80% of machine-learning work that isn't training — cleaning, engineering, shipping. In our benchmarks it gets from a raw CSV to a deployed model 7× faster than manual Jupyter, while catching 16 of 20 synthetic data flaws (sklearn catches 3). This booklet walks the why, the how, what's inside, what we measured, and how we built it.

# Agentic AutoML Platform

*From dataset to deployed models, agentially and autonomously.*

|         |                                 |
|---------|---------------------------------|
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| COURSE  | CSE 449 · Senior Design Project |
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# 01

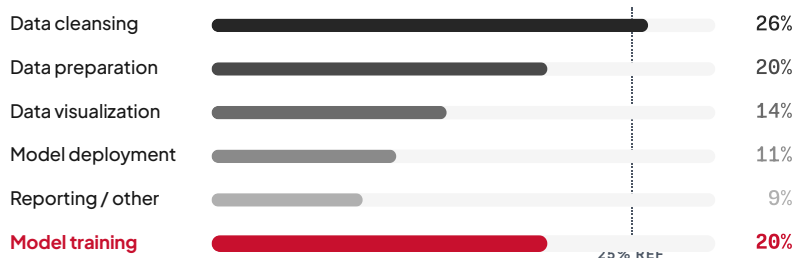
## *WHY*

data science is mostly not data science

3D DIORAMA SLOT • DIV-01-WHY.SVG

# The 80% Problem.

*Data scientists spend 80% of their time on everything except training models.*



Before a single `fit()` call, a practitioner has cleaned, merged, relabeled, visualized, sanity-checked, re-cleaned, prepped splits, and debugged feature pipelines. Training itself is a thin slice at the top of the stack.

The same survey that gives us the 80% figure breaks the other 80 into six buckets: cleansing, prep, viz, deployment, reporting, and the glue between them. None of those tasks are the idea behind the model — they're the scaffolding around it.

The pattern has been stable for a decade. It's what makes 'doing data science' feel so different from 'learning data science.'

SOURCE • ANACONDA STATE OF DATA SCIENCE 2022 • N = 3,493

# Three things unlocked at once.

## 01 / MODELS *LLMs finally read tabular schemas well.*

Models finally read tabular schemas with function calling, typed JSON, and reliable tool-call fidelity — a 2024 unlock, not a 2022 one.

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## 02 / PROTOCOLS *MCP gives agents a durable tool contract.*

Protocols give agents a durable tool contract. MCP standardizes how an agent sees the product's capabilities, logs every call, and enforces allowlists per stage.

---

## 03 / RUNTIME *Sandboxed Python is fast and safe enough now.*

Runtime is now fast and safe enough. Containerized Jupyter kernels start in under a second; a bad cell fails inside the sandbox instead of trashing the notebook.

### BY THE STACK

GPT-4o-mini: tool-call fidelity  
MCP registry: 20+ tools  
Docker: 2GB RAM • 1 CPU  
Jupyter Kernel Gateway: < 1s

# Notebook-as-ritual → agent-as-colleague.

## BEFORE

- 01 A 40-cell notebook, rewritten for every dataset.
- 02 Copy-paste from Stack Overflow and past projects.
- 03 Silent data flaws — dates parsed as integers, row indices treated as features.
- 04 No record of what was tried, or why it was rejected.
- 05 Deployment lives in a separate codebase the data scientist has never opened.

## WITH AAMT

- 01 An agent plans the workflow in natural language, in your project's context.
- 02 It writes the cells itself, citing the pandas / sklearn docs it used.
- 03 16/20 guardrail score vs sklearn's 3/20 — we catch 8 of 10 seeded flaws to sklearn's partial 1.
- 04 Every proposal, approval, and revision is captured as a durable run artifact.
- 05 The champion model ships as a downloadable notebook and serving endpoint.

## PENDING APPROVAL

*Your call: which workflow do you want to be in?*

# 02

## *HOW*

talk to it · plan with it · own the notebook

3D DIORAMA SLOT · DIV-02-HOW.SVG



# Three ways to work the notebook.

## CHAT

*Talk to your data like a colleague.*

Ask for a profile, a feature idea, or a model recommendation. The agent reads your dataset, drafts the cells, and explains its reasoning in the same pane.

→ [ASK IN ENGLISH](#)

## PLAN

*Plan with it, not around it.*

Every multi-step task surfaces as a plan card — a numbered list of proposed cells. You approve, revise, or reject before anything runs.

→ [APPROVE OR REVISE](#)

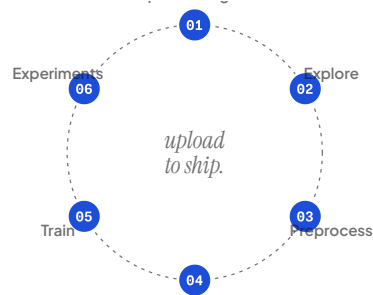
## NOTEBOOK

*Own the notebook that ships.*

Approved cells land in a real Jupyter notebook you can edit, re-run, or download. There's no hidden agent state — the notebook is the record.

→ [EDIT AND SHIP](#)

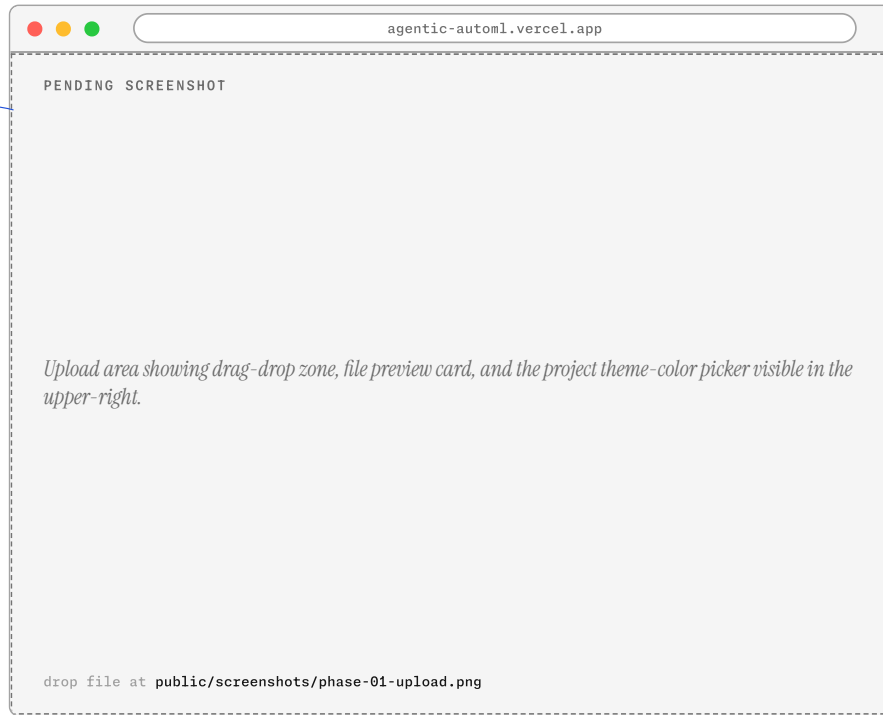
## THE 6-PHASE LOOP



# Upload & Ingest

*Drop a CSV or pull from Postgres, name the project, pick a theme.*

CSV·JSON·XLSX up to 300 MB.



Profile samples 20 rows before Postgres commits.

Theme color propagates to every visible surface.

*Ingested. Continue to Explore →*

# Explore

*Browse schema, inspect distributions, and run natural-language SQL.*

Schema panel folds/unfolds by column family.



NL → SQL box compiles to real SQL you can copy.

Two-pass planner: plan JSON, then read-only SELECT.

*Mapped. Continue to Preprocess →*

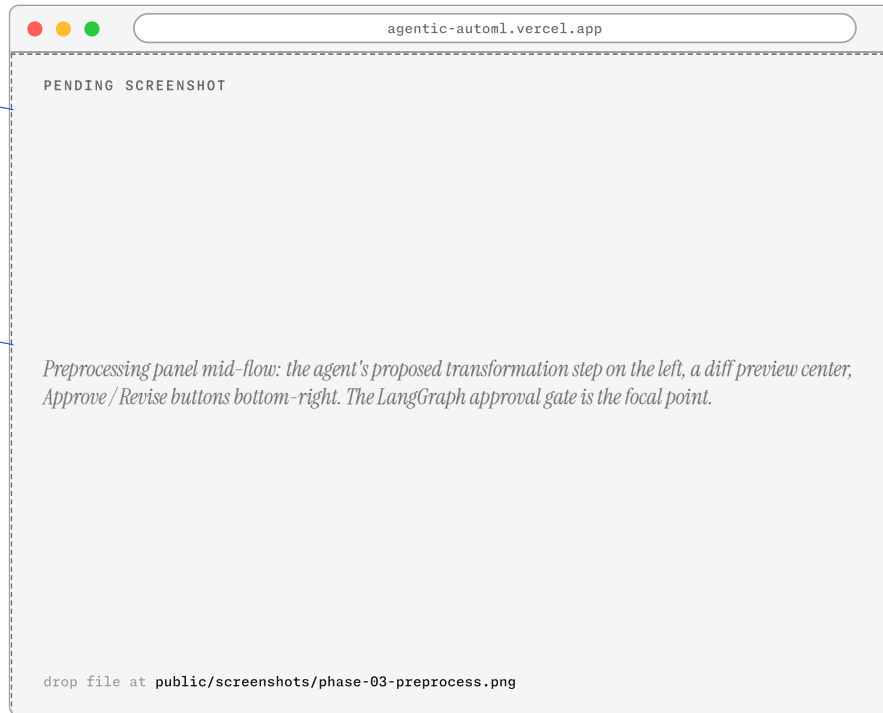
# Preprocess

*Agent proposes transforms; you approve, revise, or reject each one.*

*Each step is a reviewable diff, not a black box.*

*Auto-repair retries twice before handing back to the planner.*

*Approval gate pauses the FSM until you decide.*

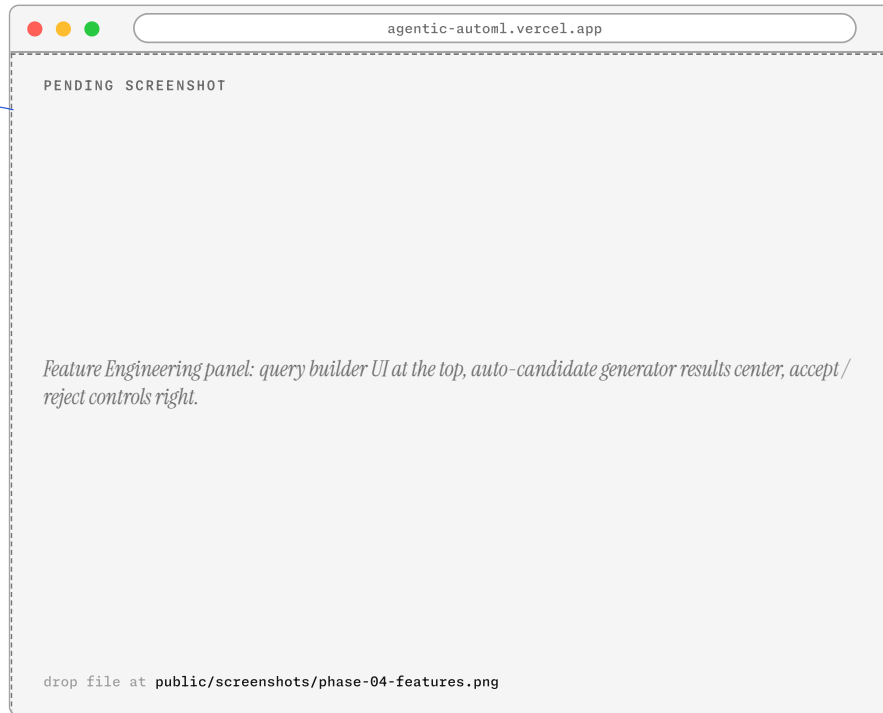


*Cleaned. Continue to Features →*

# Features

*Build features via the query UI or accept auto-generated candidates.*

*Query builder is a first-class UI, not a prompt hack.*



*Agent proposes 3-5 diverse candidates per turn.*

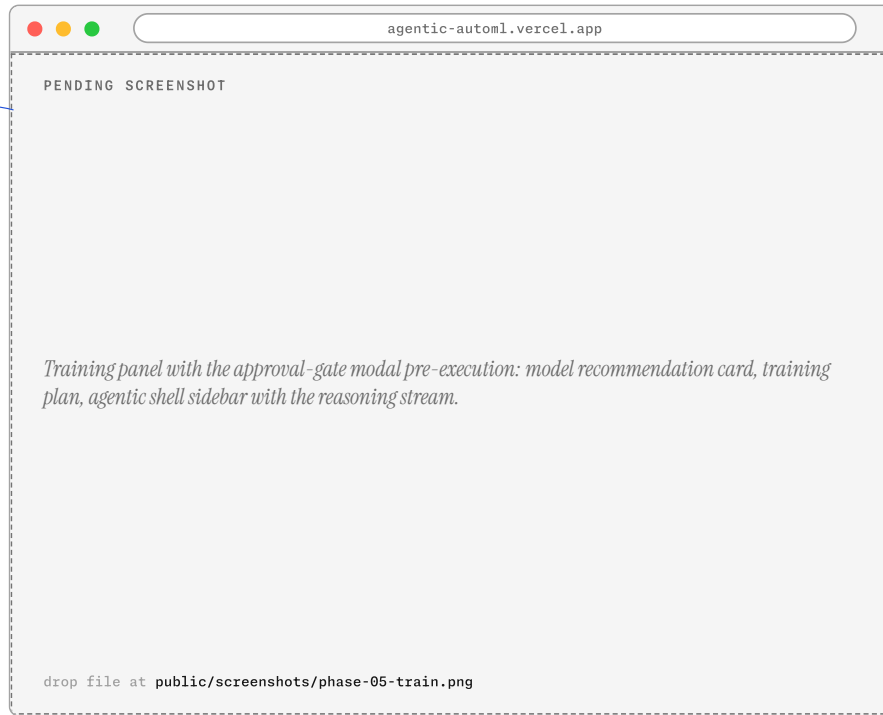
*Each tagged high · med · low impact; accept individually or in bulk.*

*Engineered. Continue to Train →*

# Train

*Approve the training plan, watch the agent's reasoning stream.*

*Default is GradientBoosting via  
sklearn.ensemble.*



*Lifecycle has two user gates: plan, then  
pre-fit review.*



*Reasoning stream stays audited after  
the run.*

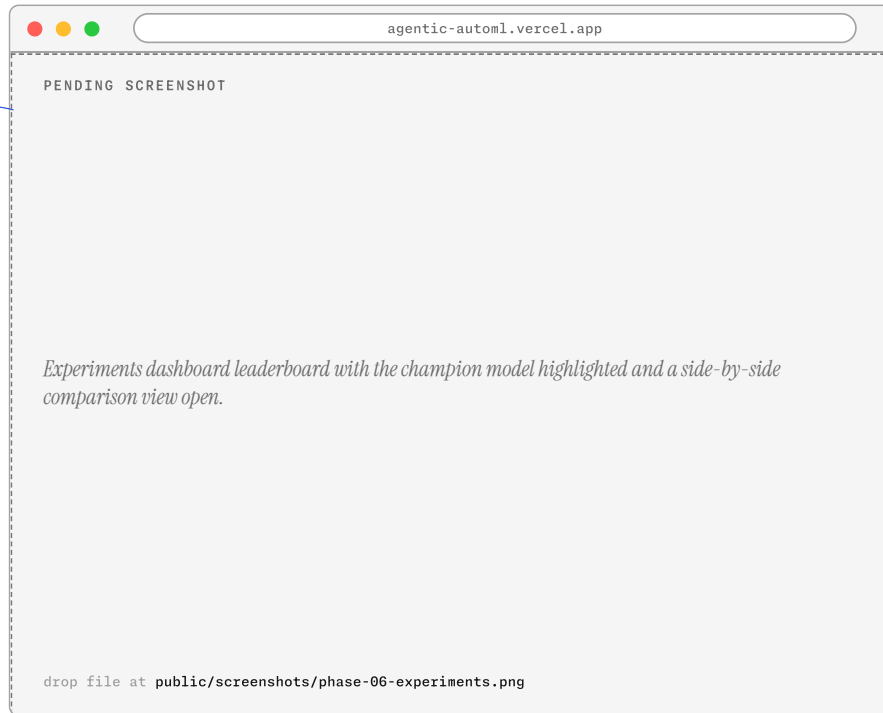


*Trained. Continue to Experiments →*

# Experiments

*Leaderboard with champion highlighted; compare runs side-by-side.*

*Welch's t-test compares model variants.*



*Leaderboard surfaces top-3; champion row always pinned.*

*Download notebook + serving endpoint per run.*

*Shipped. Now look under the hood →*

# 03

## *INSIDE*

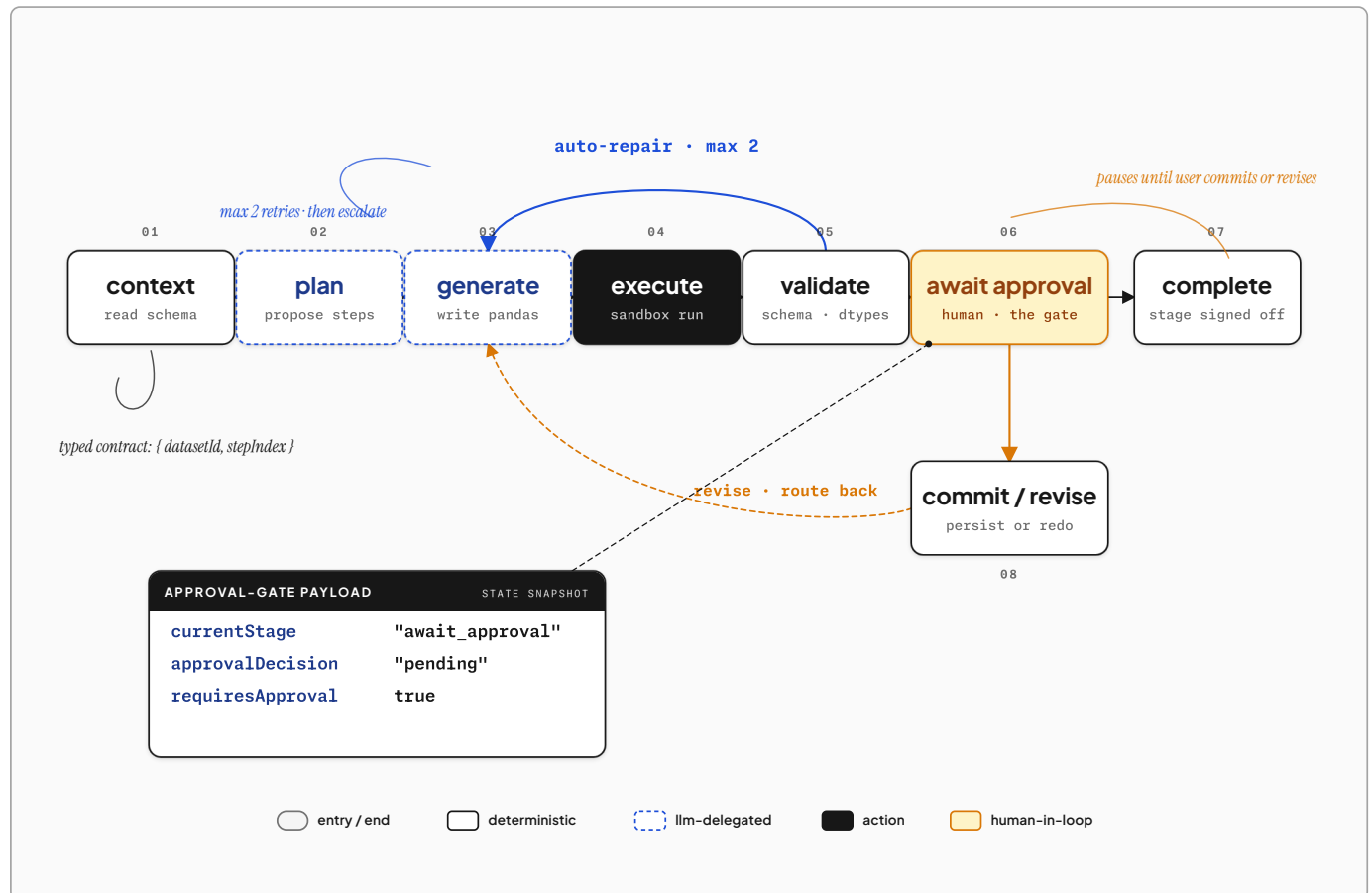
the engine room

3D DIORAMA SLOT • DIV-03-INSIDE.SVG



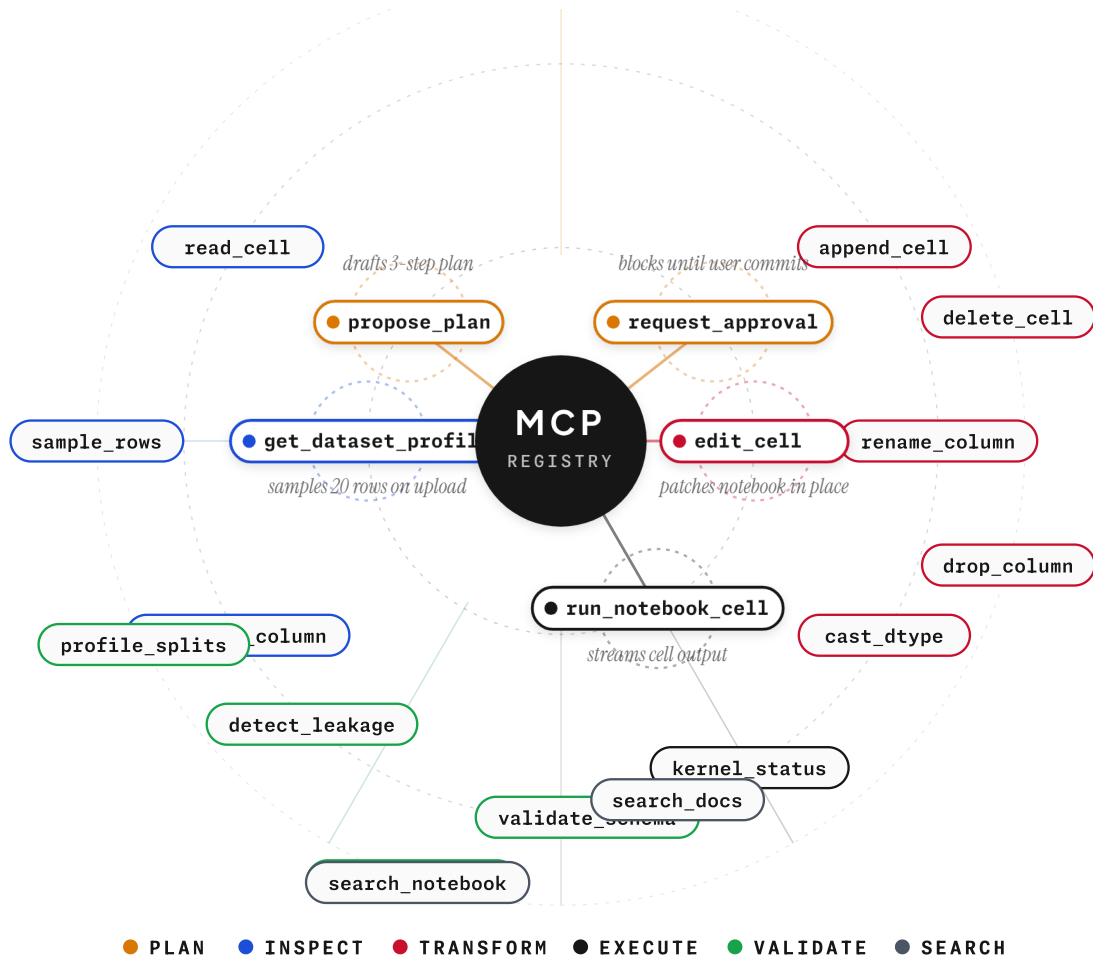
# The preprocessing FSM.

Preprocessing is an 8-stage LangGraph state machine. Each stage is a typed node with an input contract and an explicit successor set. The approval gate lives at stage 6 — the FSM pauses until the user commits or revises. Auto-repair is a bounded retry loop; after two failed attempts, control returns to the planner.



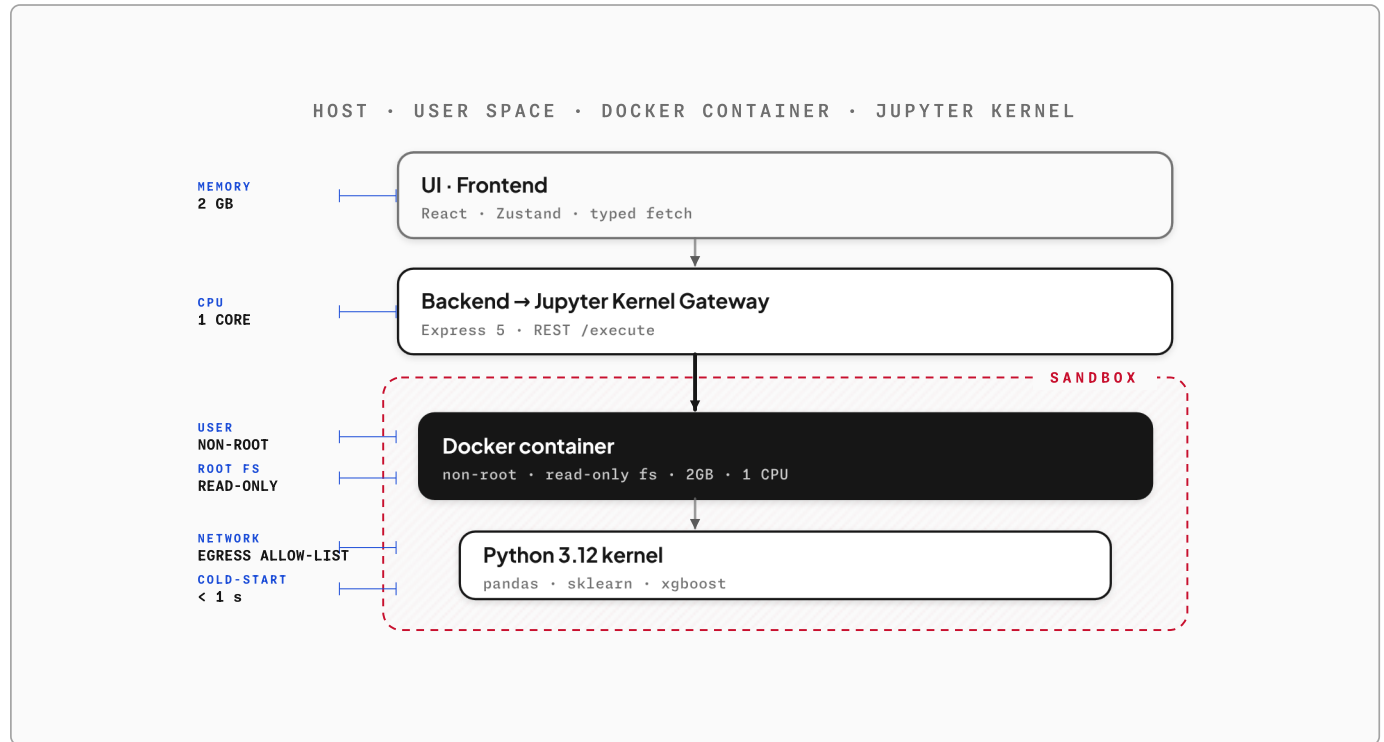
# The MCP tool registry.

20+ tools expose the product to the agent through the Model Context Protocol. Five see the most use: `get_dataset_profile` (inspect), `edit_cell` (transform), `run_notebook_cell` (execute), `propose_plan` (plan), and `request_approval` (gate). Every call is logged against the run.



# Sandbox & kernel.

Every cell runs inside a Docker container with hard limits: 2GB RAM, 1 CPU, non-root user, read-only root filesystem. A Jupyter Kernel Gateway is the only channel into the container; the gateway's REST surface is the agent's sole execution path. If the container dies, the run fails cleanly — nothing is left behind.



## PENDING APPROVAL

*Every cell of generated code is yours to read, edit, or reject before it executes.*

# 04

## *PROOF*

what we measured · how it landed

3D DIORAMA SLOT · DIV-04-PROOF.SVG

# Time-to-model, measured.

7x

*faster than manual Jupyter*

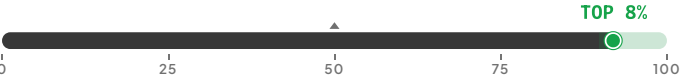
N = 25 RUNS • 5 DATASETS • GPT-4O-MINI  
• SEED = 42



*Enough time to test three more feature ideas – per session.*

# Top-tier placement with caught flaws.

QUALITY • TOP 15% ON EVERY KAGGLE BENCHMARK



HELD-OUT TEST • 5 REPS • 80/20 STRATIFIED • SEED = 42

|                           |                              |
|---------------------------|------------------------------|
| TITANIC<br>.812<br>TOP 8% | AMES<br>.122<br>TOP 12%      |
| FRAUD<br>.978<br>TOP 15%  | SPACESHIP<br>.806<br>TOP 10% |
| ADULT<br>.873<br>TOP 14%  |                              |

GUARDRAILS • 16 / 20 CAUGHT

16 points / 20 possible · 8 of 10 flaws caught, weighted.

| DATA FLAW             | OURS        | SKLEARN |
|-----------------------|-------------|---------|
| Target leakage        | ●           | ●       |
| Hidden missing values | ●           | ●       |
| Datetime as numeric   | ●           | ●       |
| Row index as feature  | ●           | ●       |
| Extreme outliers      | ●           | ●       |
| OURS                  | <div></div> |         |
| SKLEARN               | <div></div> |         |
|                       |             | 16/20   |
|                       |             | 3/20    |

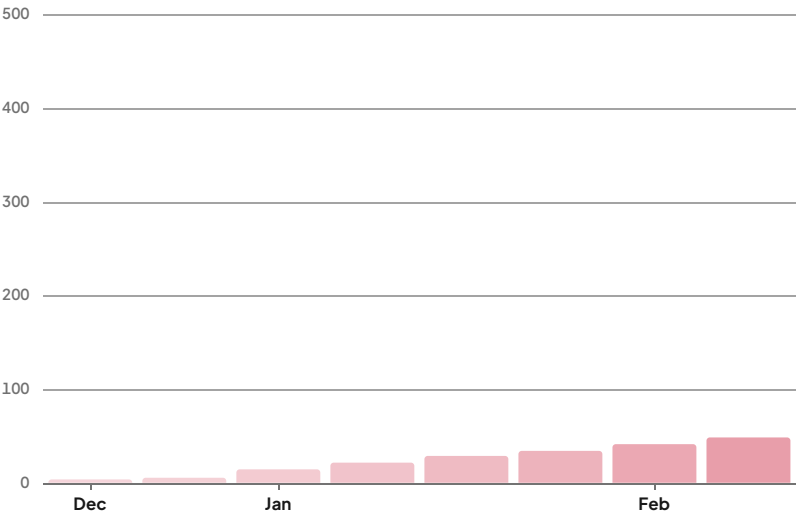
10 SEEDED FLAWS • 2 PTS EACH • SCORED ON TOOL INTENT

# 05

## *BUILD*

eleven months · two engineers · 1,989 commits

3D DIORAMA SLOT · DIV-05-BUILD.SVG



S6

DEC 21 - JAN 24

01

Project skeleton: backend · frontend · Postgres.

02

First upload → explore loop.

03

Seeded the `projects.json` store.

BY SHREE + AYUSH

DEC

"Brain and body. We split the system in half."

S7

JAN 25 - FEB 21

01

LangGraph FSM wired end-to-end.

02

MCP tool registry, 12 tools at launch.

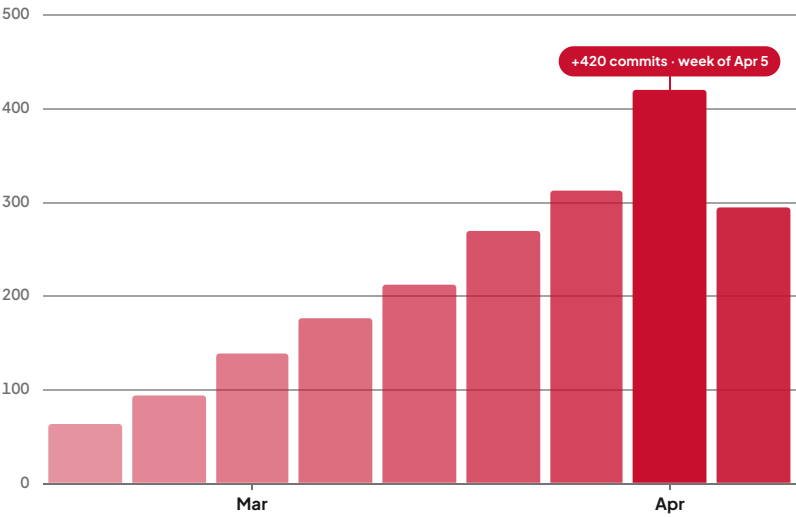
03

First approval gate shipped.

BY SHREE + AYUSH

FEB





S8

FEB 22 - MAR 21

01 Solo sprint: 205 commits.

02 Training panel + experiments dashboard.

03 Benchmark harness: 5 datasets · 25 runs.

BY SHREE

2,123 COMMITS · 324 ISSUES · 153 MRS · 61

FEB

S9

MAR 22 - APR 20

01 Peak week: +420 commits (Apr 5).

02 Landing site + poster + booklet.

03 Kaggle benchmark: TOP 15% on every leaderboard.

BY SHREE + AYUSH

ACTIVE DAYS · 11 MONTHS

APR

"*Sprint 8 was solo: 205 commits.*"

# Two engineers. One system.

*Shree Chaturvedi*

Strategy Consultant · CS + Math

01 · agent orchestration (LangGraph)  
02 · MCP tool registry  
03 · frontend experience layer

*Ayush Yadav*

Data Integration Intern · CS

01 · preprocessing pipeline  
02 · sandbox + kernel gateway  
03 · dataset ingestion

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TECHNICAL ADVISOR

*Samer Y. Khamaiseh, Ph.D.*

CSE 449 STEWARD

*Prof. Lynn Stahr, M.S.*

*Thanks to Dr. Khamaiseh (technical advisor) and Prof. Stahr (CSE 449 steward).*

Built on LangGraph, Jupyter Kernel Gateway, scikit-learn, and pandas.

*“Brain and body. We split the system in half.”*

END

# Try it.

*From dataset to deployed models, agentially and autonomously.*

LIVE DEMO

[agentic-automl.vercel.app](https://agentic-automl.vercel.app)

 *try it*

REPO

[gitlab.csi.miamioh.edu/.../ai-augmented-auto-ml-toolchain](https://gitlab.csi.miamioh.edu/.../ai-augmented-auto-ml-toolchain)

*read it* 



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ART/COVER-BACK.SVG